

Midterm Assessment Answer Sheet

Semester 1, 2023/2024

Time allowed: 2 hours

Instructions (please read carefully):

- Write down your **Student Number** on the right and using ink or pencil, **shade completely** the corresponding bubbles in the grid for each digit or letter. **DO NOT WRITE YOUR NAME!**
- This answer booklet comprises **EIGHTEEN (18) pages**, including this cover page.
- This is a **OPEN-SHEET** assessment. You are allowed one A4-sized double-sided cheatsheet.
- Weightage of questions is given in square brackets. The maximum attainable score is 100.
- You are allowed to bring a calculator, but it cannot have any form of external communication capability, i.e. not Wifi- or 4G-enabled. Mobile phones and tablets are not allowed.
- All questions must be answered in the space provided on the answer sheet; no extra sheets will be accepted as answers.
- You are allowed to write with pencils, as long as it is legible.
- Marks may be deducted** for unrecognisable handwriting and/or for not shading the student number properly.
- You must submit only the **ANSWER SHEET** and no other documents. The question set may be used as scratch paper.
- An excerpt of the question may be provided to aid you in answering in the correct box, where applicable. It is not the exact question. You should still refer to the original question in the question booklet.

STUDENT NUMBER														
A														
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For Examiner's Use Only

Question	Marks
Q1	/ 32
Q2	/ 26
Q3	/ 24
Q4	/ 15
Q5	/ 3
Total	/100

Question 1A State representation.

[2 marks]

Question 1B Invariant.

[2 marks]

Question 1C Initial and goal state.

[2 marks]

Question 1D Actions.

[2 marks]

Question 1E Best search algorithm.

[4 marks]

Question 1F Use Graph-Search or not?

[3 marks]

Question 1G Initial candidate solution.

[2 marks]

Question 1H Transition function.

[2 marks]

Question 1I Heuristic function.

[2 marks]

Question 1J New formulation for Variant #1.

[6 marks]

Question 1K Best search algorithm for Variant #1.

[2 marks]

Question 1L Is your TREE-SEARCH complete and/or optimal? Is the search tree finite for Variant #1?

[3 marks]

Question 2A Is h_A admissible? Explain.

[3 marks]

☐ Admissible ☐ Not admissible

Question 2B Is h_B admissible? Explain.

[3 marks]

☐ Admissible ☐ Not admissible

Question 2C Is h_C admissible? Explain.

[3 marks]

☐ Admissible ☐ Not admissible

Question 2D Is h_D admissible and/or consistent? Explain.

[6 marks]

☐ Admissible ☐ Not admissible
☐ Consistent ☐ Not consistent

Question 2E Is h_E admissible and/or consistent? Explain.

[6 marks]

- ☐ Admissible

☐ Not admissible

☐ Consistent

☐ Not consistent

Question 2F Choosing between h_E and h_F ? Explain.

[5 marks]

Question 3A The eligibility of a low income loan taker?

[2 marks]

- ☐ Yes

☐ No

☐ Undefined

Question 3B What is the entropy?

[2 marks]

- ☐ $0.0 \leq \text{Entropy} < 0.2$

☐ $0.2 \leq \text{Entropy} < 0.4$

☐ $0.4 \leq \text{Entropy} < 0.6$

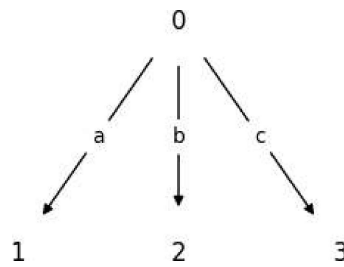
☐ $0.6 \leq \text{Entropy} < 0.8$

☐ $0.8 \leq \text{Entropy} \leq 1.0$

Show your working below (if you wish):

Question 3C One-level Tree.

[6 marks]



What is node 0?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☐ - |

What is node 1?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☐ - |

What is node 2?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☐ - |

What is node 3?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☐ - |

What is edge a?

- | | | | | | |
|---------------------------------|------------------------------|----------------------------|-----------------------------|-----------------------------------|--------------------------|
| ☐ excellent | ☐ fair | ☐ high | ☐ low | ☐ middle-aged | ☐ no |
| ☐ poor | ☐ senior | ☐ yes | ☐ young | ☐ - | |

What is edge b?

- | | | | | | |
|---------------------------------|------------------------------|----------------------------|-----------------------------|-----------------------------------|--------------------------|
| ☐ excellent | ☐ fair | ☐ high | ☐ low | ☐ middle-aged | ☐ no |
| ☐ poor | ☐ senior | ☐ yes | ☐ young | ☐ - | |

What is edge c?

- | | | | | | |
|---------------------------------|------------------------------|----------------------------|-----------------------------|-----------------------------------|--------------------------|
| ☐ excellent | ☐ fair | ☐ high | ☐ low | ☐ middle-aged | ☐ no |
| ☐ poor | ☐ senior | ☐ yes | ☐ young | ☐ - | |

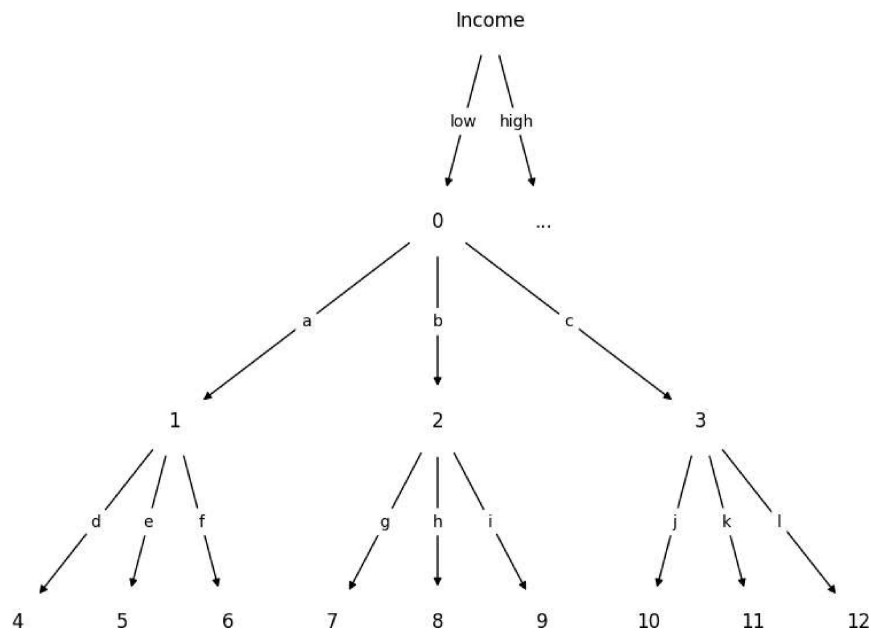
Question 3C One-level Tree. (Continued)

[6 marks]

Show your working below (if you wish):

Question 3D Full Decision Tree.

[6 marks]



What is node 0?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|--------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ no |
| ☐ yes | ☐ no/yes | ☐ - | |

What is node 1?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|--------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ no |
| ☐ yes | ☐ no/yes | ☐ - | |

What is node 2?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|--------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ no |
| ☐ yes | ☐ no/yes | ☐ - | |

What is node 3?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|--------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ no |
| ☐ yes | ☐ no/yes | ☐ - | |

What is node 4?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|--------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ no |
| ☐ yes | ☐ no/yes | ☐ - | |

What is node 5?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|--------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ no |
| ☐ yes | ☐ no/yes | ☐ - | |

What is node 6?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|--------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ no |
| ☐ yes | ☐ no/yes | ☐ - | |

Are there any nodes from 7 onwards?

- | | |
|---------------------------|--------------------------|
| ☐ Yes | ☐ No |
|---------------------------|--------------------------|

Question 3D Full Decision Tree. (Continued)

[6 marks]

What is edge a?

- | | | | | |
|---------------------------------|----------------------------|-----------------------------------|--------------------------|----------------------------|
| ☐ excellent | ☐ fair | ☐ middle-aged | ☐ no | ☐ poor |
| ☐ senior | ☐ yes | ☐ young | ☐ - | |

What is edge b?

- | | | | | |
|---------------------------------|----------------------------|-----------------------------------|--------------------------|----------------------------|
| ☐ excellent | ☐ fair | ☐ middle-aged | ☐ no | ☐ poor |
| ☐ senior | ☐ yes | ☐ young | ☐ - | |

What is edge c?

- | | | | | |
|---------------------------------|----------------------------|-----------------------------------|--------------------------|----------------------------|
| ☐ excellent | ☐ fair | ☐ middle-aged | ☐ no | ☐ poor |
| ☐ senior | ☐ yes | ☐ young | ☐ - | |

What is edge d?

- | | | | | |
|---------------------------------|----------------------------|-----------------------------------|--------------------------|----------------------------|
| ☐ excellent | ☐ fair | ☐ middle-aged | ☐ no | ☐ poor |
| ☐ senior | ☐ yes | ☐ young | ☐ - | |

What is edge e?

- | | | | | |
|---------------------------------|----------------------------|-----------------------------------|--------------------------|----------------------------|
| ☐ excellent | ☐ fair | ☐ middle-aged | ☐ no | ☐ poor |
| ☐ senior | ☐ yes | ☐ young | ☐ - | |

What is edge f?

- | | | | | |
|---------------------------------|----------------------------|-----------------------------------|--------------------------|----------------------------|
| ☐ excellent | ☐ fair | ☐ middle-aged | ☐ no | ☐ poor |
| ☐ senior | ☐ yes | ☐ young | ☐ - | |

Question 3D Full Decision Tree. (Continued)

[6 marks]

Show your working below (if you wish):

Question 3E Precision, Recall, F1.

[5 marks]

Number of True Positives (TP)?

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ >4

Number of False Positives (FP)?

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ >4

Number of True Negatives(TN)?

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ >4

Number of False Negatives (FN)?

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ >4

Precision?

- ☐ $0.0 \leq \text{Precision} < 0.2$ ☐ $0.2 \leq \text{Precision} < 0.4$
☐ $0.4 \leq \text{Precision} < 0.6$ ☐ $0.6 \leq \text{Precision} < 0.8$
☐ $0.8 \leq \text{Precision} \leq 1.0$

Recall?

- ☐ $0.0 \leq \text{Recall} < 0.2$ ☐ $0.2 \leq \text{Recall} < 0.4$
☐ $0.4 \leq \text{Recall} < 0.6$ ☐ $0.6 \leq \text{Recall} < 0.8$
☐ $0.8 \leq \text{Recall} \leq 1.0$

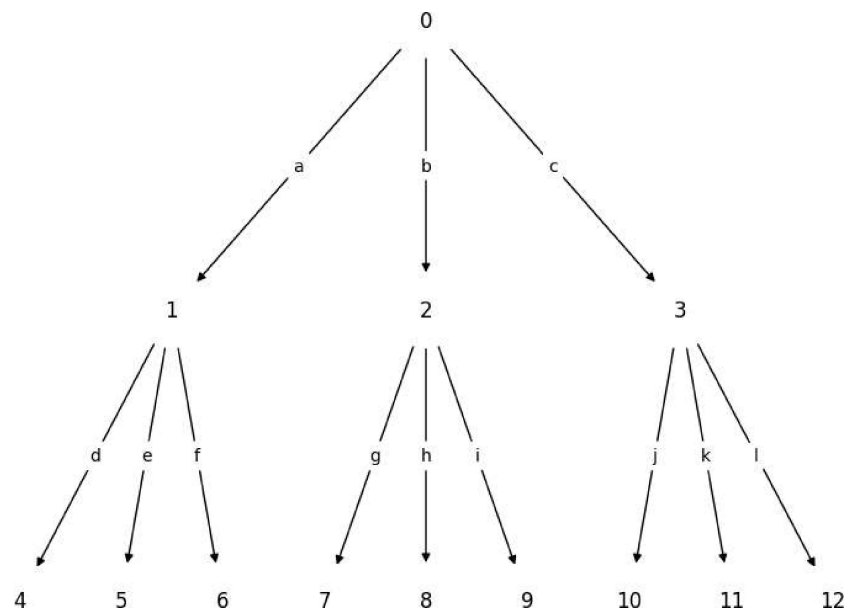
F1 Score?

- ☐ $0.0 \leq \text{F1 Score} < 0.2$ ☐ $0.2 \leq \text{F1 Score} < 0.4$
☐ $0.4 \leq \text{F1 Score} < 0.6$ ☐ $0.6 \leq \text{F1 Score} < 0.8$
☐ $0.8 \leq \text{F1 Score} \leq 1.0$

Show your working below (if you wish):

Question 3F Min-Sample DT.

[3 marks]



What is node 0?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☐ - |

What is node 1?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☐ - |

What is node 2?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☐ - |

What is node 3?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☐ - |

What is node 4?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☐ - |

What is node 5?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☐ - |

What is node 6?

- | | | | |
|---------------------------|-------------------------------------|---------------------------------------|------------------------------|
| ☐ Age | ☐ Credit Rating | ☐ Criminal Record | ☐ Income |
| ☐ no | ☐ yes | ☐ no/yes | ☐ - |

Are there any nodes from 7 onwards?

- | | |
|---------------------------|--------------------------|
| ☐ Yes | ☐ No |
|---------------------------|--------------------------|

Question 3F Min-Sample DT. (Continued)

[3 marks]

What is edge a?

- | | | | | | |
|---------------------------------|------------------------------|----------------------------|-----------------------------|-----------------------------------|--------------------------|
| ☐ excellent | ☐ fair | ☐ high | ☐ low | ☐ middle-aged | ☐ no |
| ☐ poor | ☐ senior | ☐ yes | ☐ young | ☐ - | |

What is edge b?

- | | | | | | |
|---------------------------------|------------------------------|----------------------------|-----------------------------|-----------------------------------|--------------------------|
| ☐ excellent | ☐ fair | ☐ high | ☐ low | ☐ middle-aged | ☐ no |
| ☐ poor | ☐ senior | ☐ yes | ☐ young | ☐ - | |

What is edge c?

- | | | | | | |
|---------------------------------|------------------------------|----------------------------|-----------------------------|-----------------------------------|--------------------------|
| ☐ excellent | ☐ fair | ☐ high | ☐ low | ☐ middle-aged | ☐ no |
| ☐ poor | ☐ senior | ☐ yes | ☐ young | ☐ - | |

What is edge d?

- | | | | | | |
|---------------------------------|------------------------------|----------------------------|-----------------------------|-----------------------------------|--------------------------|
| ☐ excellent | ☐ fair | ☐ high | ☐ low | ☐ middle-aged | ☐ no |
| ☐ poor | ☐ senior | ☐ yes | ☐ young | ☐ - | |

What is edge e?

- | | | | | | |
|---------------------------------|------------------------------|----------------------------|-----------------------------|-----------------------------------|--------------------------|
| ☐ excellent | ☐ fair | ☐ high | ☐ low | ☐ middle-aged | ☐ no |
| ☐ poor | ☐ senior | ☐ yes | ☐ young | ☐ - | |

What is edge f?

- | | | | | | |
|---------------------------------|------------------------------|----------------------------|-----------------------------|-----------------------------------|--------------------------|
| ☐ excellent | ☐ fair | ☐ high | ☐ low | ☐ middle-aged | ☐ no |
| ☐ poor | ☐ senior | ☐ yes | ☐ young | ☐ - | |

Show your working below (if you wish):

Question 4A Game Tree.

[6 marks]

What is the minimum number of variables required to maintain the game state?

- ☐ 1 ☐ 2 ☐ 3 ☐ 7 ☐ 8 ☐ Others

How many nodes (not including the leaves) are there in the game tree?

- ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7
☐ 8 ☐ 9 ☐ 10 ☐ 11 ☐ 12 ☐ 13 ☐ 14
☐ 15 ☐ 16 ☐ 17 ☐ 18 ☐ 19 ☐ 20 ☐ >20

What is the number of leaves where the first player wins?

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ >6

What is the number of leaves where the first player loses?

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ >6

What is the total number of edges in the game tree?

- ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7
☐ 8 ☐ 9 ☐ 10 ☐ 11 ☐ 12 ☐ 13 ☐ 14
☐ 15 ☐ 16 ☐ 17 ☐ 18 ☐ 19 ☐ 20 ☐ >20

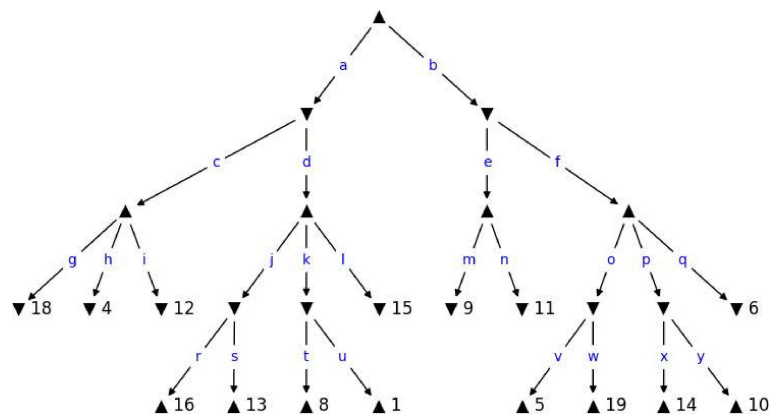
Which player do we expect to win?

- ☐ First player ☐ Second player ☐ It depends ☐ Draw

Show your working below (if you wish):

Question 4B Alpha-beta left-to-right.

[5 marks]



What is the root value?

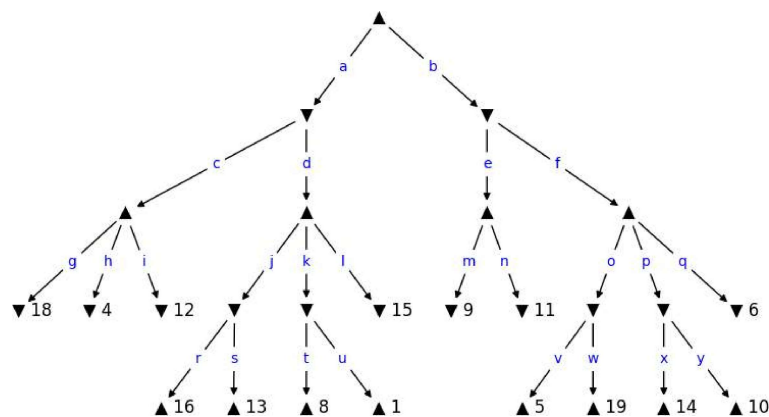
- ☐ 1 ☐ 4 ☐ 5 ☐ 6 ☐ 8 ☐ 9 ☐ 10 ☐ 11
☐ 12 ☐ 13 ☐ 14 ☐ 15 ☐ 16 ☐ 18 ☐ 19

Which of the following link(s) are pruned? Shade all that is/are true.

- ☐ a ☐ b ☐ c ☐ d ☐ e ☐ f ☐ g
☐ h ☐ i ☐ j ☐ k ☐ l ☐ m ☐ n
☐ o ☐ p ☐ q ☐ r ☐ s ☐ t ☐ u
☐ v ☐ w ☐ x ☐ y

Question 4C Alpha-beta right-to-left.

[4 marks]

Which of the following link(s) are pruned? Shade all that is/are true.

- ☐ a ☐ b ☐ c ☐ d ☐ e ☐ f ☐ g
☐ h ☐ i ☐ j ☐ k ☐ l ☐ m ☐ n
☐ o ☐ p ☐ q ☐ r ☐ s ☐ t ☐ u
☐ v ☐ w ☐ x ☐ y

Question 5 3 most important lessons

[3 marks]

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